Assignment 5

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Set Up

Packages

```
#general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(scales)
library(NatParksPalettes)
```

Data

```
# taxnomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)
```

Necessary Code from Class

```r
# creating new object from sites
ncos_sites <- sites %>%
  # filering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")



# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))


# creatinga a new object from water quality
water_quality_clean <- water_quality %>%
  # cleaning column names
  clean_names() %>%
```

```
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") %>%
  # creating new column to join with later data frame
  unite("date_site", date, site, remove = TRUE) %>%
  # grouping by date/site
  group_by(date_site) %>%
  # calculating median pH, salinity, and DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
  # ungrouping the data frame
  ungroup() %>%
  # filtering out salinity outliers
  filter(date_site != "2022-08-12NVBR")

# creating a new object from aq_ins
aq_ins_clean <- aq_ins %>%
  # cleaning column names
  clean_names() %>%
  # filtering join: only include quarterly NCOS surveys from ncos_site
  semi_join(ncos_sites, by = "site") %>%
  # renaming column
  rename(date = date_on_vial) %>%
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         annelida_polychaete,
         diptera_chironomid) %>%
  # filtering sample types
  filter(sample_type %in% c("FB250", "FB 250")) %>%
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
```

```r
  ) %>%
  group_by(date, site, taxon) %>%
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  # ungrouping the data frame
  ungroup() %>%
  # created new column to join with data frame
  unite("date_site", date, site, remove = FALSE) %>%
  # joined with water quality clean
  left_join(water_quality_clean, by = "date_site") %>%
  # joined wtth taxanomic information
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  # making a new column called site_full with proper names
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NDC" ~ "Devereux Creek")) %>%
  # reordering site full by median salinity
  mutate(site_full = fct_reorder(site_full,
                                 med_sal,
                                 .fun = "median",
                                 .na_rm = TRUE),
         # reversing the order of site_full
         site_full = fct_rev(site_full))

# creating new object from aq_ins_clean
copepods <- aq_ins_clean %>%
  # filtering to only include copepod observations
  filter(taxon == "copepod") %>%
  # creating a new column that takes the natural log of average abundance per
  ↪  liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

1. Visualizing differences across sites in major taxon

a. Planning the Figure

- x-axis: Salinity (ppt)

- y-axis: taxon
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

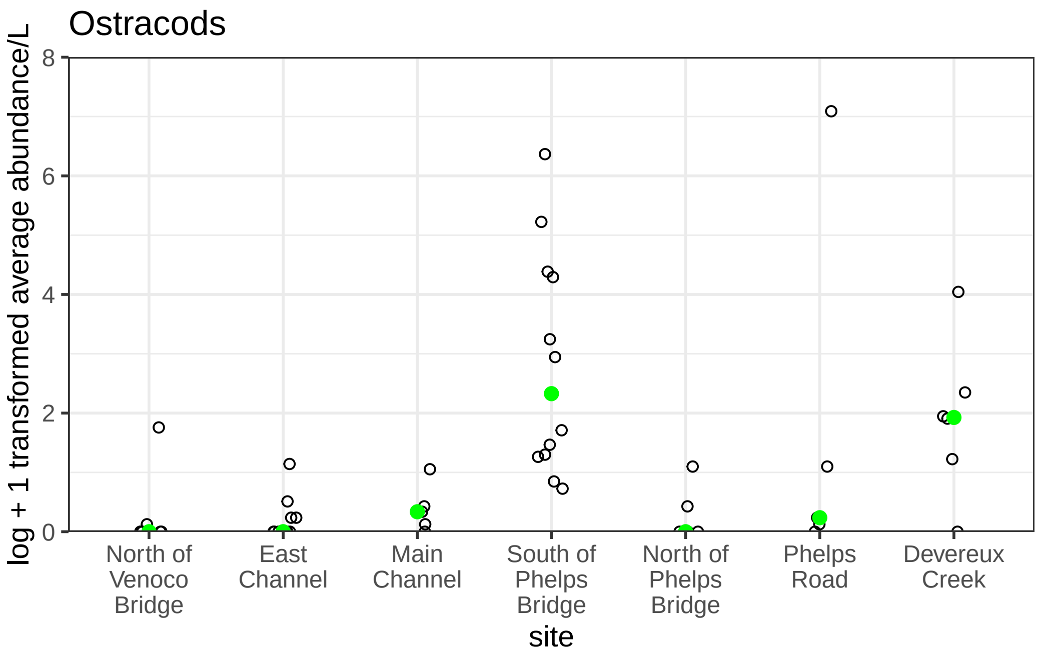
b. Wrangling the data

```
# creating a new object called ostracods
ostracods <- aq_ins_clean %>%
  # filtering to only include ostracods
  filter(taxon == "ostracod") %>%
  # creating a new column that takes the natural log of average abundance per
  ↪  liter
  mutate(log_ave_lit = log(ave_lit + 1))

# taking a random sample from the new data frame
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-05-12_NPB2 2022-05-12 00:00:00 NPB2  ostr~ 1.20e+3  NA      NA     NA
 2 2022-08-15_NMC  2022-08-15 00:00:00 NMC   ostr~ 1.33e-1   7.61   NA     NA
 3 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1  ostr~ 0        NA      NA     NA
 4 2023-05-04_NEC  2023-05-04 00:00:00 NEC   ostr~ 6.67e-1   8.05   NA      8.55
 5 2022-05-06_NMC  2022-05-06 00:00:00 NMC   ostr~ 0         9.96    1.45  16.5
 6 2023-02-21_NEC  2023-02-21 00:00:00 NEC   ostr~ 0         8.56   NA      8.8
 7 2023-05-05_NPB  2023-05-05 00:00:00 NPB   ostr~ 2.47e+1  NA      NA     12.1
 8 2023-11-14_NPB  2023-11-14 00:00:00 NPB   ostr~ 1.8 e+1  NA      NA     NA
 9 2022-08-10_NPB  2022-08-10 00:00:00 NPB   ostr~ 2.67e+0   8.28    8.84   9.81
10 2024-01-28_NPB  2024-01-28 00:00:00 NPB   ostr~ 3.33e+0   7.43    3.08   6.6
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Coding the figure

```r
# creating a new object that stores the ostracod ggplot
ostracod_plot <- ggplot(data = ostracods,
                       mapping = aes(x = site_full,
                                     y = log_ave_lit))+
  # plotting ostracod observations with geom_jitter
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # adding a distinct green median point
  stat_summary(geom = "point",
              fun = median,
              color = "green",
              size = 2)+
  # getting rid of overlapping labels
  scale_x_discrete(labels = label_wrap(width = 10)) +
  # getting rid of y-axis space at the bottom of the graph
  scale_y_continuous(expand = c(0, 0),
                     limits = c(0, 8)) +
  # adding labels
  labs(x = "site",
        y = "log + 1 transformed average abundance/L",
        title = "Ostracods") +
  # getting off the default theme
  theme_bw()

ostracod_plot
```

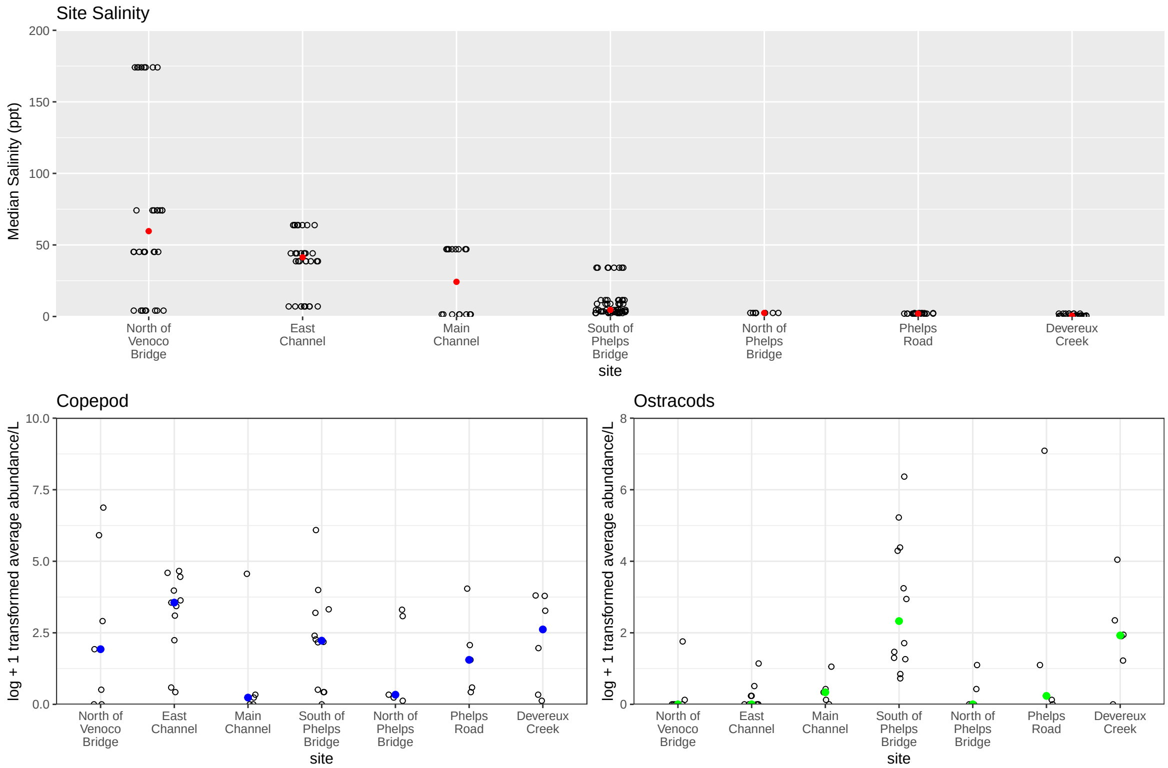


d. Creating a multipanel figure

```
# creating an object of a salinity plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal))+
  # plotting salinity measurements with geom_jitter
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21)+
  # adding a distinct red median point
  stat_summary(geom = "point",
               fun = median,
               color = 'red')+
  # getting rid of overlapping labels
  scale_x_discrete(labels = label_wrap(width = 10))+
  # getting rid of y-axis space at the bottom
  scale_y_continuous(expand = c(0, 0),
                     limits = c(0, 200)) +
  # adding labels
  labs(x = "site",
```

```r
       y = " Median Salinity (ppt)",
       title = "Site Salinity")



# creating a plotting object of copepod abundance
copepod_plot <- ggplot(data = copepods,
                       mapping = aes(x = site_full,
                                     y = log_ave_lit))+
  # plotting copepod observations using geom_jitter
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # adding a distinct blue median point
  stat_summary(geom = "point",
              fun = median,
              color = "blue",
              size = 2)+
  # getting rid of overlapping labels
  scale_x_discrete(labels = label_wrap(width = 10)) +
  # getting rid of y-axis space at the bottom
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 10)) +
  # adding labels
  labs(x = "site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod") +
  # getting off the default theme
  theme_bw()
```

```r
# printing a multi panel view of my 3 plots
salinity_plot /
  (copepod_plot | ostracod_plot)
```



e. Interpreting the figure

The biggest differences in copepod and ostracod abundance across sites can be seen at the East Channel site, where copepods are highest in abundance, and osrtacods are extremely low in abundance. The two species are similar in abundance at the Main Channel, South of Phelps Bridge, North of Phelps Bridge, and Devereux Creek sites.

There doesn't appear to be a correlation between site salinity and copepod or ostracod abundance. Median salinity is order by descending ppt from left to right on the x-axis, and the invertebrate abundance plots with the same x-axis don't follow a descending or ascending pattern, appearing relatively random.

2. Visualizing total abundance as a function of salinity

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: log_avg_lit

- color: taxon
- geometry: geom_points()
- facets: facet_wrap(~ taxon)

b. Wrangling the Data

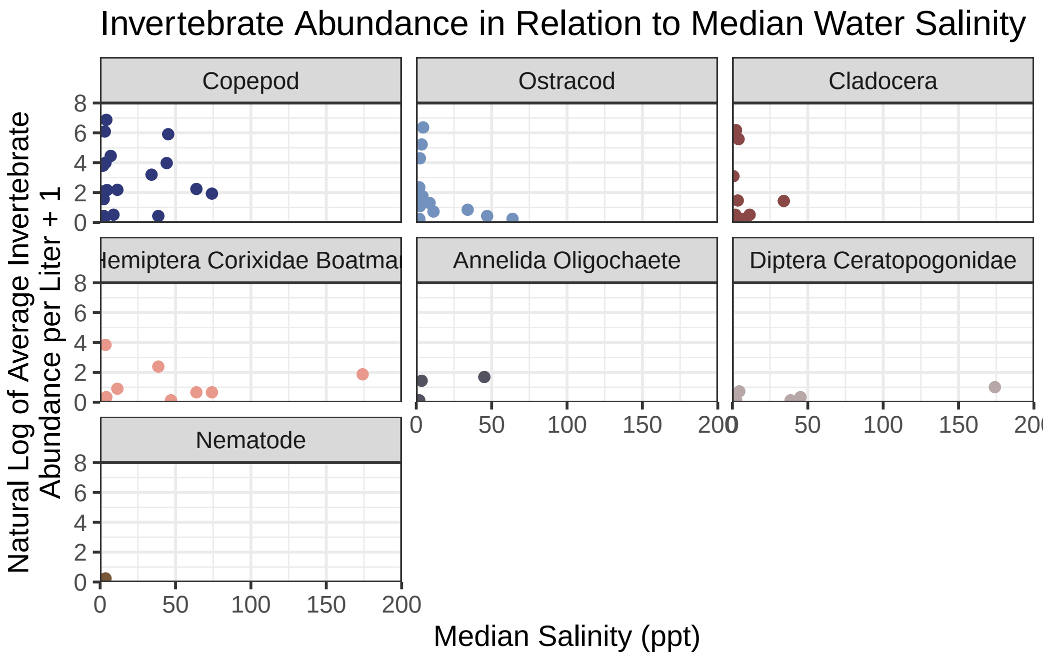
```
# creating a new object from aq_ins_clean
avg_taxon_abundance <- aq_ins_clean %>%
  # filtering out na salinity rows
  filter(!is.na(med_sal)) %>%
  # filtering out rows with 0 invertebrate observations, better for
   ↪  visualization
  filter(!ave_lit == 0) %>%
  # selecting relevant columns
  select(date_site, taxon, med_sal, ave_lit) %>%
  # creating new columns, correcting taxon column
  mutate(
    # converting taxon to title case
    taxon = to_any_case(taxon, case = "title"),
    # taking natural log of average abundance per liter
    log_ave_lit = log(ave_lit + 1),
    # reorder taxon column based on max average abundance
    taxon = fct_reorder(taxon, ave_lit, .fun = "max", .na_rm = TRUE),
    # reversing order so that top rows appear first
    taxon = fct_rev(taxon))

# printing a radnom sample of 10 rows
slice_sample(avg_taxon_abundance, n = 10)
```

```
# A tibble: 10 x 5
   date_site       taxon     med_sal ave_lit log_ave_lit
   <chr>           <fct>       <dbl>   <dbl>       <dbl>
 1 2022-05-12_NPB  Cladocera  11.4      0.667       0.511
 2 2021-11-23_NPB2 Copepod     1.99     6.93        2.07
 3 2022-05-11_NDC  Cladocera   0.398    0.133       0.125
 4 2020-01-21_NDC  Ostracod    0.3      2.4         1.22
 5 2022-01-28_NDC  Ostracod    2.00     9.47        2.35
 6 2022-08-08_NPB2 Cladocera   2.36   484           6.18
 7 2022-02-23_NVBR Copepod    45.2    368.          5.91
 8 2024-01-28_NPB  Ostracod    3.08     3.33        1.47
```

```
 9 2023-03-03_NPB2 Copepod     1.98     0.533       0.427
10 2023-03-03_NPB2 Ostracod    1.98     0.133       0.125
```

c. Coding the figure

```r
# plotting avg_taxon_abundance with salinity on the x-axis and average
↪  abundance on the y-axis
ggplot(data = avg_taxon_abundance,
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon))+
  # plotting
  geom_point()+
  # making a new panel for each taxon
  facet_wrap(~ taxon)+
  # getting off the default theme
  theme_bw()+
  # adding custom color palette
  scale_color_manual(values = natparks.pals("Torres"))+
  # removing the legend
  theme(legend.position = "none")+
  # removing space on the y-axis at the bottom
  scale_y_continuous(expand = c(0,0),
                     limits = c(0,8))+
  # removing space on the x-axis
  scale_x_continuous(expand = c(0,0),
                     limits = c(0,200))+
  # adding labels
  labs(x = "Median Salinity (ppt)",
       y = "Natural Log of Average Invertebrate\nAbundance per Liter + 1",
       title = "Invertebrate Abundance in Relation to Median Water Salinity")
```



d. Interpretting the Figure

All taxa appear to prefer water with salinity between 0 and 100 ppt, as most data points are clustered within that range. The invertebrates that are most abundant at relatively low salinity levels (0-50 ppt) are Copepods, Ostracods, and Cladocera, who all spike in abundance, with Copepods nearly reaching a maximum of of 966 individuals per liter at only 4.1 median salinity. There are two notable exceptions to the trend of high abundance in low salinity - Hemiptera Corixidae Boatman and Diptera Ceratopogonidae were both observed at fairly high abundance on August 8th 2022 when median salinity was at 174 ppt, while all others species were not observed. Overall, this figure shows that invertebrates are found in higher levels of abundance when salinity is low.